

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Operated and maintained professional 3D printers and desktop scanning hardware in production environments at HCL Technologies, optimising print files, performing post-processing (cleaning, curing, finishing, inspection), and conducting hands-on reverse engineering workflows including mesh cleanup, alignment, and CAD recreation using **SolidWorks** and enterprise CAD tools.
- Managed additive manufacturing production support and incident response for enterprise customers, troubleshooting hardware and software issues, advising on printer consumables and Bambu Lab printer selection, and maintaining detailed **application notes and best-practice documentation** for cross-functional teams and customer knowledge bases.
- Performed technical pre-sales support and customer demonstrations, translating manufacturing requirements into 3D printing and scanning solutions, providing application engineering advice on integrating digital manufacturing into customer workflows, and quoting suitable equipment and materials for real-world use cases.
- Conducted reverse engineering and 3D scanning demonstrations using handheld and desktop scanners, supporting customers through part production workflows, performing basic to intermediate mesh and CAD recreation, and documenting workflows and technical specifications for post-implementation support.
- Developed **Python automation scripts and data pipelines** for production monitoring and quality reporting, applying logical, detail-oriented problem-solving to validate manufacturing data, coordinate change-control processes, and ensure regulatory compliance across interconnected enterprise systems.
- Maintained clean, organised, and safe laboratory environments while managing hardware inventory, consumables, and equipment calibration; answered customer support calls and provided troubleshooting guidance with strong communication skills and confidence engaging with customers during technical demonstrations and site visits.

SKILLS

3D Printing & Additive Manufacturing – 3D printer operation and maintenance, desktop and professional printers, Bambu Lab printer selection and quoting, print file optimisation, post-processing (cleaning, curing, finishing, inspection), consumables management, production support.

3D Scanning & Reverse Engineering – handheld and desktop scanner operation, 3D scanning demonstrations, mesh cleanup and alignment, CAD recreation, reverse engineering workflows, point-cloud processing.

CAD & Design Software – SolidWorks, CAD recreation, digital manufacturing workflows, tolerance and materials specification, design documentation.

Engineering & Manufacturing – materials science, manufacturing processes, tolerances, additive manufacturing production workflows, quality inspection, change-control management, lab environment management.

Technical Support & Customer Engagement – application engineering, pre-sales technical input, customer troubleshooting, technical demonstrations, customer advisory and workflow integration, documentation and training materials.

Data & Automation – Python scripting, production monitoring, quality reporting, data pipeline development, SQL, Linux.

Professional Skills – logical problem-solving, detail-oriented methodology, technical communication, customer service, driver's license (full and clean).

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

3D Printing & Reverse Engineering Production Dashboard (HCL Technologies)

- Designed and deployed an enterprise **production monitoring dashboard** for managing 3D printer queues, part inventory, and reverse engineering project workflows using **Power BI** and **SQL data pipelines**, enabling real-time visibility into manufacturing progress and consumables usage across multiple customer jobs.
- Automated print-file validation and post-processing workflows using **Python scripts**, reducing manual quality-check time by **35%** and improving consistency of finished-part inspection across desktop and professional printers.
- Documented additive manufacturing best practices, customer application workflows, and reverse engineering procedures in centralised knowledge base, enabling new team members to ramp up quickly and supporting consistent customer advisory and technical demonstrations.

Manufacturing Systems Integration & Change Management (HCL Technologies)

- Integrated 3D printing equipment management, inventory tracking, and customer project-order systems via **REST APIs** and **database-level data flows**, ensuring seamless handoffs between production scheduling, quality reporting, and customer notifications.
- Coordinated change-control management and UAT processes for new hardware deployments, equipment firmware updates, and software tool upgrades, working cross-functionally with operations teams, customer stakeholders, and engineering to validate functionality and ensure zero-downtime transitions.
- Built **application notes and troubleshooting guides** documenting common hardware issues, optimal print parameters, and reverse engineering workflows, ensuring customer support teams and field engineers could rapidly diagnose and resolve technical problems with confidence.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)